

Unit 4 Polynomials, Remainder and Factor Theorem	
1.	(a) The cubic polynomial $f(x)$ is such that the roots of $f(x) = 0$ are $x = 2, x = -1$ and $x = k$ and the coefficient of x^3 is 1. Calculate the value of k if $f(x)$ leaves a remainder of -128 when divided by $(x - 3)$. (b) Given that $2x^3 - 4x^2 - px - 2 \equiv (x - 1)(x + 2)Q(x) + 5x + q$, where $Q(x)$ is the quotient. (i) State the degree of $Q(x)$. (ii) Find the value of p and of q. (c) A sum of money $\\$(3x^3 - 5x^2 + 6x - 54)$ is to be divided equally among $(x - 3)$ grandchildren. Find the share of money, in terms of x, each grandchild will receive. Ans: (a) 35 (b)(i) 1 (ii) $p = 15; q = -14$ (c) $\\$(3x^2 + 4x + 18)$
2.	(a) Given that $f(x) = ax^3 + bx^2 + 2$ leaves a remainder of -2 when divided by $(x + 2)$ and $3f(x) + 6$ is exactly divisible by $(x - 1)$. Find the value of a and of b. (b) (i) Factorise $6x^3 - 7x^2 - 8x + 5$ completely. (ii) Hence, solve the equation $5x^3 - 8x^2 - 7x + 6 = 0$. Ans: (a) $a = -1, b = -3$ (b)(i) $(x + 1)(2x - 1)(3x - 5)$ (ii) $x = -1, x = 2, x = \frac{3}{5}$
3.	The term containing the highest power of x in the polynomial $f(x)$ is $2x^3$. Two of the roots of the equation $f(x) = 0$ are -4 and 2. It is given that $f(x)$ leaves a remainder of 5 when divided by $x + 3$. (i) Express $f(x)$ in descending powers of x. (ii) Find the remainder when $f(x)$ is divided by $2x - 1$. Ans: (i) $f(x) = 2x^3 + 9x^2 - 6x - 40$ (ii) -40
4.	The cubic polynomial $f(x)$ is such that the coefficient of x^3 is 3 and its constant term is -8. The roots of the equation $f(x) = 0$ are $-\frac{1}{3}, p$ and q. Given that $f(x)$ has a remainder of -20 when divided by $x - 1$ and p is a positive integer, find (i) the value of p and of q, (ii) the remainder when $f(x)$ is divided by $2x - 1$.

	Ans: (i) $p = 2, q = -4$ (ii) $-16\frac{7}{8}$
5.	(i) Find the value of h for which $x^3 + hx^2 + 4hx - (3 + h)$ is divisible by $x + 1$. (ii) Using the value of h found in part (i), solve the equation $x^3 + hx^2 + 4hx - (3 + h) = 0$, expressing non-integer roots in the form $c \pm \sqrt{d}$, where c and d are integers. Ans: (i) -1 (ii) -1 or $1 \pm \sqrt{3}$
6.	(i) The expression $f(x) = abx^3 + (a - 2b)x^2 - 4ax - (a + b)$, where $a \neq 0$ and $b \neq 0$, is exactly divisible by $x^2 - 4$. Find the value of a and b. (ii) Hence factorise $f(x)$ completely. (iii) Find the remainder when $f(x)$ is divided by $(2x - 1)$. Ans: (i) $a = 3, b = 1$ (ii) $f(x) = (x + 2)(x - 2)(3x + 1)$ (iii) Remainder $= -9\frac{3}{8}$
7.	The expression $px^3 - x^2 + qx + r$, where $p > 0$, leaves a remainder of -14 when divided by $x + 2$, and has a factor of $x^2 + x$. Find the values of p, q and r. Ans: $p = 2, q = -3, r = 0$
8.	The function f is defined by $f(x) = 2x^3 + 4ax^2 - 6x^2 + (10 - 15a)x + 11a - 6$, where a is a constant. (i) Show that $x - 1$ is a factor of $f(x)$. (ii) Find the possible values of a for which the equation $f(x) = 0$ only has exactly 2 real roots. Ans: (ii) $a = -4, a = 0.5$
9.	The expression $f(x) = 5x^3 + ax^2 + b$ where a and b are constants, has a factor of $x + 1$ and leaves a remainder of 51 when divided by $x - 2$. (i) Find the value of a and of b. (ii) Using the values of a and b found in part (i) determine the number of real roots of the equation $5x^3 + ax^2 + b = 0$, justifying your answer. Ans: (i) $a = 2, b = 3$ (ii) one real root

10.	The term containing the highest power of x in the polynomial $f(x)$ is $2x^4$ and the roots of $f(x) = 0$ are 2 and -7. $f(x)$ has a remainder of -72 when divided by $(x + 1)$, and a remainder of -80 when divided by $(x - 1)$. (i) Find the expression for $f(x)$ in descending power of x. (ii) Explain why the equation $f(x) = 0$ has exactly 2 real roots. Ans: (i) $f(x) = 2x^4 + 13x^3 - 8x^2 - 17x - 70$
11.	The expression $2x^3 + ax^2 + bx - 9$, where a and b are constants, has a factor $x - 3$ and leaves a remainder of 8 when divided by $x + 1$. (i) Find the value of a and of b. (ii) Using the values of a and b found in part (i), factorise the expression $2x^3 + ax^2 + bx - 9$ completely. Hence solve the equation $16x^3 + 4ax^2 = 9 - 2bx$. Ans: (i) $a = 1, b = -18$ (ii) $f(x) = (x - 3)(2x + 1)(x + 3); x = \frac{3}{2}, -\frac{1}{4}, -\frac{3}{2}$
12.	Given that $x + 5$ is a common factor of $x^3 + kx^2 - Ax + 15$ and $x^3 - x^2 - (A + 5)x + 40$, find the value of k and of A. Hence factorise $x^3 + kx^2 - Ax + 15$ completely. Ans: $A = 17, k = 1; (x + 5)(x - 3)(x - 1)$
13.	It is given that $f(x) = x^3 + ax^2 + bx + 14$ leaves a remainder of 5 when divided by $x^2 - 9$. (i) Find the values of a and of b. (ii) Using your values of a and b, solve the equation $f(x) = 0$. Ans: (i) $a = -1, b = -9$ (ii) 2, 2.19 or -3.19
14.	The polynomial $P(x) = 2x^3 + ax^2 + bx + 8$, where a and b are constants, leaves a remainder of 10 when divided by $2x - 1$. Given that $x + 2$ is a factor of $P(x)$. (i) Find the value of a and of b (ii) Explain why the equation $P(x) = 0$ has only 1 real root. Hence find this root.